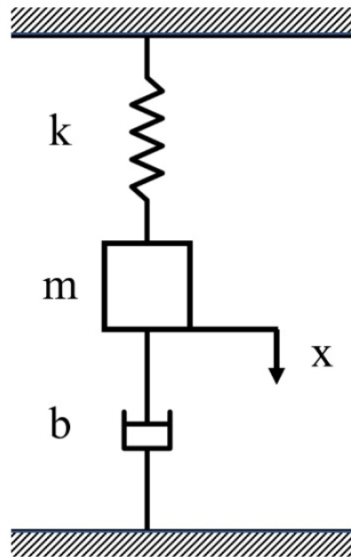


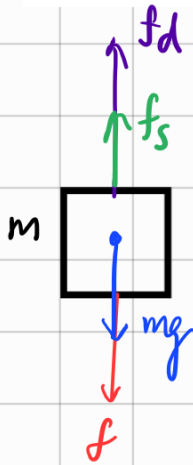
1. Problem 1

Derive the equation of motion (EOM) in the following figure. Note that the external force/external moment applied in this question acts on the mass and in the direction of the arrow.

1.1 a



FBD:



Spring stiffness (k)
Damping coefficient (b)
Mass (m)

$$f_s = kx \quad f_d = b \cdot v = b\dot{x}$$

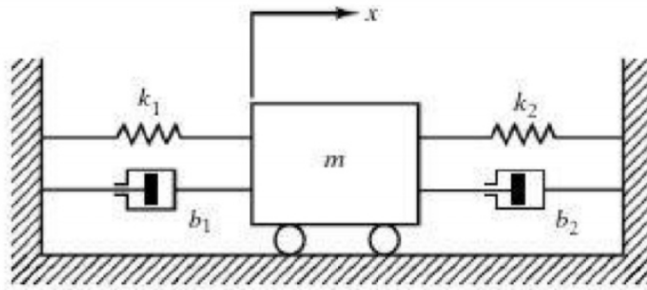
$$m\ddot{x} = -kx - b\dot{x} + mg + f$$

$$m\ddot{x} + b\dot{x} + kx = mg + f$$

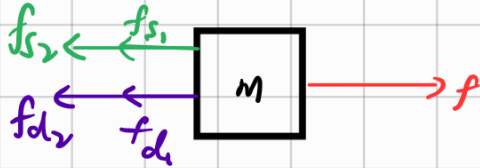
$$\underline{m\ddot{x}_u + b\dot{x}_u + kx_u = mg + f} \quad - \text{undeformed position}$$

$$\underline{m\ddot{x}_d + b\dot{x}_d + kx_d = f} \quad - \text{deformed position}$$

1.2 b



FBD:



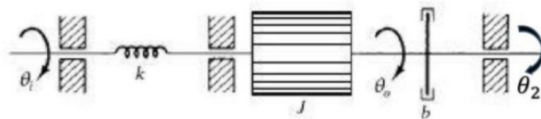
$$f_{s1} = k_1 x \quad f_{s2} = k_2 x$$

$$f_{d1} = b_1 \dot{x} \quad f_{d2} = b_2 \dot{x}$$

$$m\ddot{x} = -k_1 x - k_2 x - b_1 \dot{x} - b_2 \dot{x} + f$$

$$\underline{m\ddot{x} + (b_1 + b_2)\dot{x} + (k_1 + k_2)x = f}$$

1.3 c



Angle in the figure are $\theta_0, \theta_1, \theta_2$. Derive the EOM for θ_0 .

J moment of inertia of the mass

Torsional spring constant (k)

Torsional damping coefficient (b)

$$\text{Spring torque } T_s = k(\theta_0 - \theta_1)$$

$$\text{Damper torque } T_d = b(\omega_2 - \omega_1) = b(\dot{\theta}_2 - \dot{\theta}_0)$$

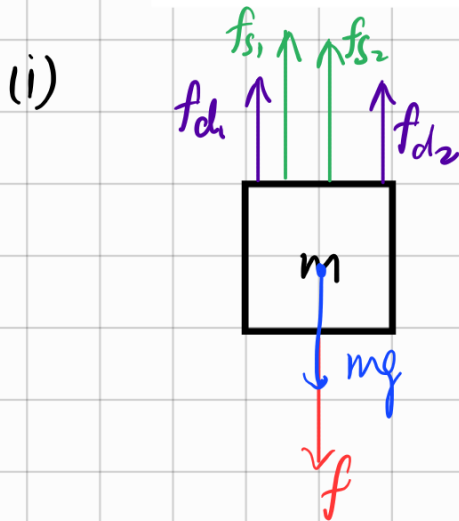
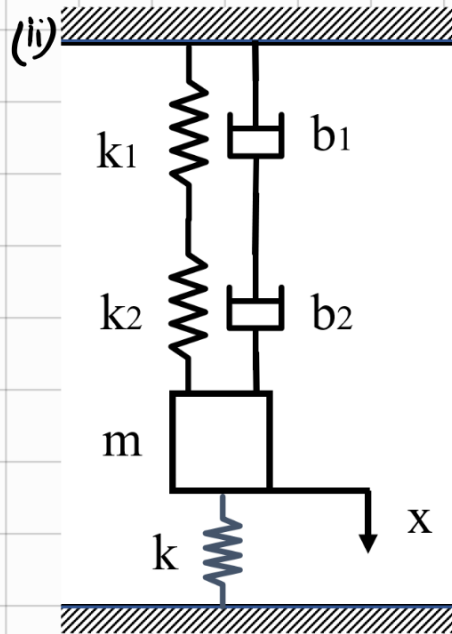
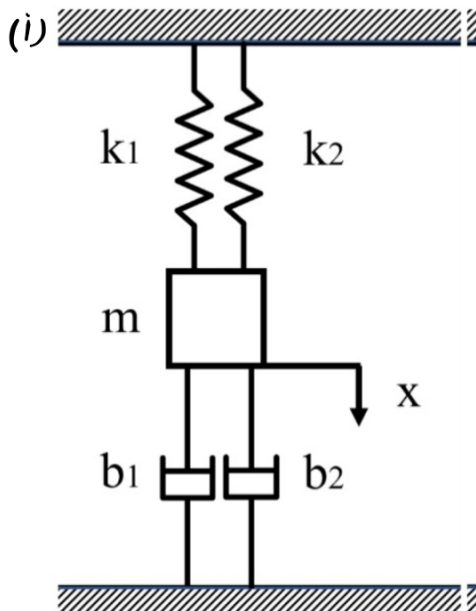
From Newton's second law for rotation:

$$J\ddot{\theta}_0 = -k(\theta_0 - \theta_1) - b(\dot{\theta}_2 - \dot{\theta}_0) + \tau$$

$$\underline{J\ddot{\theta}_0 + k(\theta_0 - \theta_1) + b(\dot{\theta}_2 - \dot{\theta}_0) = \tau}$$

2.1 a

Derive the equation represented by each figure below

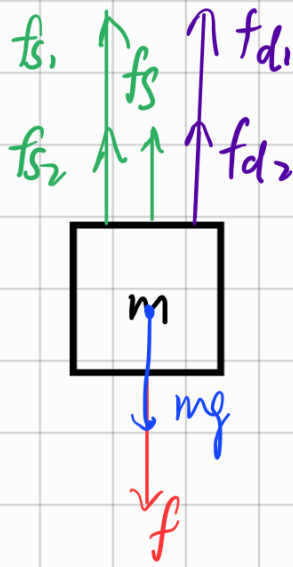


$$\begin{aligned} m\ddot{x} &= -(k_1 + k_2)x - (b_1 + b_2)\dot{x} + mg + f \\ \underline{m\ddot{x} + (k_1 + k_2)x + (b_1 + b_2)\dot{x} &= mg + f} \end{aligned}$$

Springs in parallel: $f_s = \overbrace{(k_1 + k_2)}^{k_{eq}} x$
 Dampers in parallel: $f_d = \overbrace{(b_1 + b_2)}^{b_{eq}} \dot{x}$

springs in series: $f_s = \frac{k_1 k_2}{k_1 + k_2} x$
 dampers in series: $f_d = \frac{b_1 b_2}{b_1 + b_2} \dot{x}$

(ii)



$$k_{eq} = \frac{k_1 k_2}{k_1 + k_2} + k$$

$$b_{eq} = \frac{b_1 b_2}{b_1 + b_2}$$

$$m\ddot{x} = -\left(\frac{k_1 k_2}{k_1 + k_2} + k\right)x - \left(\frac{b_1 b_2}{b_1 + b_2}\right)\dot{x} + mg + f$$

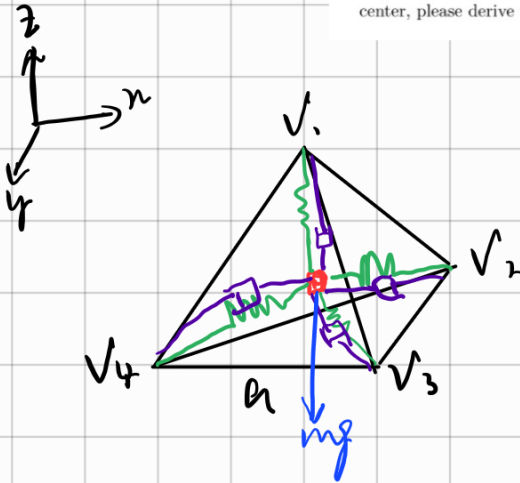
$$\underline{m\ddot{x} + \left(\frac{b_1 b_2}{b_1 + b_2}\right)\dot{x} + \left(\frac{k_1 k_2}{k_1 + k_2} + k\right)x = mg + f}$$

2.2 b (Dynamics equations expressed in vectors)

In multi-degree-of-freedom systems, we often use a state vector to represent the system's state. Here, we provide an example where the coordinates in 3D space are used as the state vector.

Assume that there is a tetrahedron with a particle of mass m located at the center of the tetrahedron. Each of the vertices of the tetrahedron is connected to the particle by a shock absorber, which can be approximated as a spring and a damper, with parameters k and b . Please derive the dynamics equation for this system, and express the resulting dynamics equation in matrix form.

If the particle is not at the center of the tetrahedron but at a position displaced by a vector $\mathbf{r}$ from the center, please derive the dynamics equation for this case.



Assuming edge of tetrahedron $= a = 1$
particle of mass at the origin $(0, 0, 0)$

the height of tetrahedron $= \frac{\sqrt{6}}{3}a$

the radius of circumsphere $R = \frac{\sqrt{6}}{4}a$

$$h_0 = \frac{\sqrt{6}}{3}a - \frac{\sqrt{6}}{4}a = \frac{\sqrt{6}}{12}a$$

$$\text{insphere } r = \frac{1}{3}a$$

$$\vec{V}_1 = (0, 0, \frac{\sqrt{6}}{3})$$

$$\vec{V}_3 = (-\frac{\sqrt{3}}{6}, -\frac{1}{2}, -\frac{\sqrt{6}}{12})$$

$$\vec{V}_2 = (\frac{\sqrt{3}}{3}, 0, -\frac{\sqrt{6}}{12})$$

$$\vec{V}_4 = (-\frac{\sqrt{3}}{6}, \frac{1}{2}, -\frac{\sqrt{6}}{12})$$



$\Rightarrow$

$$m\ddot{z} = -k_1 z - b_1 \dot{z} = mg$$

$$m \begin{pmatrix} \ddot{x} \\ \ddot{y} \\ \ddot{z} \end{pmatrix} = b \begin{pmatrix} \dot{x} \\ \dot{y} \\ \dot{z} \end{pmatrix} + k \begin{pmatrix} x \\ y \\ z \end{pmatrix} + \begin{pmatrix} 0 \\ 0 \\ mg \end{pmatrix}$$

3. Problem 3

In this problem, we will explore the dynamics equations of a particle system.

Although referred to as a particle system, this is essentially just another way of increasing the degrees of freedom. In this problem, we recommend using **generalized coordinates** to represent the system's state. We also expect students to convert the final dynamics equations into matrix form.

3.1 Tips

Using the Lagrange method can simplify the calculation.

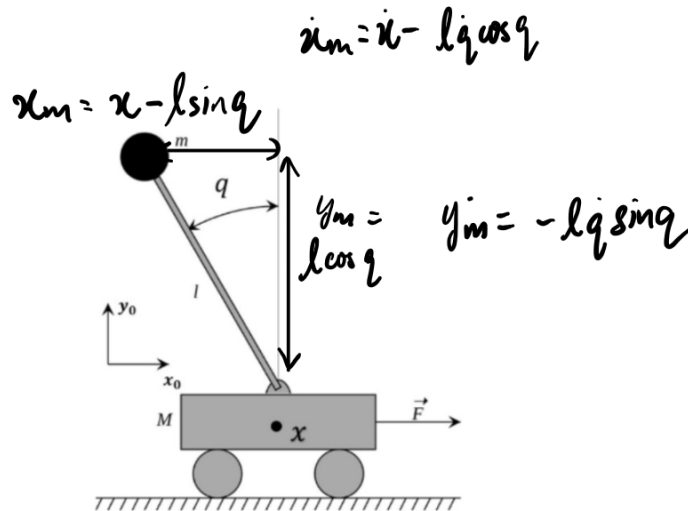
The Lagrange function of the system is

$$L = T - V$$

where T represents the kinetic energy and V represents the potential energy. Finally, the EOM can be derived from the Lagrange equation

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}_i} - \frac{\partial L}{\partial q_i} = Q_i$$

3.2 a



In the diagram, we have a first-order inverted pendulum system. Neglect the mass of the rod and all forms of friction, and derive the dynamics equation when the pendulum deviates by a **small angle**. (Note: The centrifugal force term will lead to nonlinearity in the system equation. If you wish to simplify the problem to a linear one, please ignore the centrifugal force term.)

V potential energy

T kinetic energy

$$V = mgl \cos q$$

$$\begin{aligned} T &= \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m (\dot{x}_m^2 + \dot{y}_m^2) \\ &= \frac{1}{2} M \dot{x}^2 + \frac{1}{2} m (\dot{x}^2 - 2l\dot{q}\dot{x}\cos q + l^2\dot{q}^2\cos^2 q + l^2\dot{q}^2\sin^2 q) \\ &= \frac{1}{2} (M+m) \dot{x}^2 - ml\dot{q}\dot{x}\cos q + \frac{1}{2} ml^2\dot{q}^2 \end{aligned}$$

$$L = T - V = \frac{1}{2} (M+m) \dot{x}^2 - ml\dot{q}\dot{x}\cos q + \frac{1}{2} ml^2\dot{q}^2 - mgl \cos q$$

EOM

$$\lambda: \frac{\partial L}{\partial \dot{x}} = (M+m)^2 \dot{x} - ml\dot{q}\cos q$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} = (M+m)^2 \ddot{x} - (ml\dot{q}\cos q - ml\dot{q}^2 \sin q)$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}} - \frac{\partial L}{\partial x} = (M+m)^2 \ddot{x} - (ml\dot{q}\cos q - ml\dot{q}^2 \sin q) - 0 = F(t)$$

$$q: \frac{\partial L}{\partial \dot{q}} = mgl \sin q + ml\dot{x} \sin q$$

$$\frac{\partial L}{\partial \dot{q}} = -ml\dot{x}\cos q + ml^2\dot{q}$$

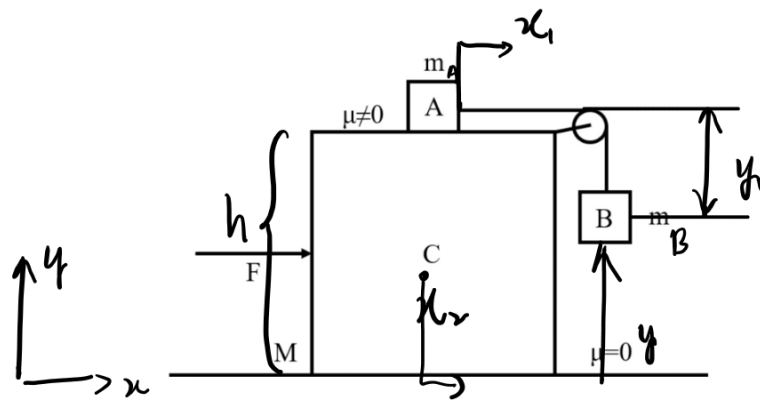
$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} = -ml\ddot{x}\cos q + ml\dot{x}\sin q + ml^2\ddot{q}$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{q}} - \frac{\partial L}{\partial q} = -ml\ddot{x}\cos q + ml\dot{x}\sin q + ml^2\ddot{q} - mgl \sin q - ml\dot{x}\sin q = 0$$

↖ no external moment

$$(\div ml) \quad \ddot{x}\cos q + \cancel{\dot{x}\sin q} + l\ddot{q} - g\sin q - \cancel{\dot{x}\sin q} = 0$$

$$\underline{\ddot{x}\cos q + l\ddot{q} - g\sin q = 0}$$



In the figure, there are three particles, A, B, and C. Please design the state variables for the system and derive the system's dynamics equations. (Note: particle B may experience oscillations due to acceleration.)

$$V = m_B g (h - y_0 - x_1) \quad T = \frac{1}{2} M_C \dot{x}_2^2 + \frac{1}{2} m_A \dot{x}_1^2 + \frac{1}{2} m_B (\dot{x}_1^2 + \dot{y}^2)$$

$$L = T - V = \frac{1}{2} M_C \dot{x}_2^2 + \frac{1}{2} m_A \dot{x}_1^2 + \frac{1}{2} m_B (\dot{x}_1^2 + \dot{y}^2) - m_B g (h - y_0 - x_1)$$

EOM

$$x_1: \frac{\partial L}{\partial x_1} = m_B g \quad \frac{\partial L}{\partial \dot{x}_1} = m_A \dot{x}_1 + m_B \dot{x}_1 \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_1} = m_A \ddot{x}_1 + m_B \ddot{x}_1$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}_1} - \frac{\partial L}{\partial x_1} = m_A \ddot{x}_1 + m_B \ddot{x}_1 - m_B g = f_k = \mu_k m_A g$$

$$\underline{m_A \ddot{x}_1 + m_B (\ddot{x}_1 - g) = \mu_k m_A g + m_B g}$$

$$x_2: \frac{\partial L}{\partial x_2} = 0 \quad \frac{\partial L}{\partial \dot{x}_2} = M_C \dot{x}_2 \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{x}_2} = M_C \ddot{x}_2$$

$$\frac{d}{dt} \frac{\partial L}{\partial \dot{x}_2} - \frac{\partial L}{\partial x_2} = \underline{M_C \ddot{x}_2 = F(t)}$$

$$y: \frac{\partial L}{\partial y} = 0 \quad \frac{\partial L}{\partial \dot{y}} = m_B \dot{y} \quad \frac{d}{dt} \frac{\partial L}{\partial \dot{y}} = m_B \ddot{y}$$

$$\underline{\frac{d}{dt} \frac{\partial L}{\partial \dot{y}} - \frac{\partial L}{\partial y} = m_B \ddot{y}}$$

4. *Problem 4 (breadth extension)

Beyond the field of dynamics, many other areas also involve system modeling and control. In this problem, we will explore system modeling in the context of biological communities. Please supplement any additional parameters as needed.

4.1 a

In a vast Qingqing grassland, there are two different biotic populations, referred to as "Xi" and "Hui". The interspecies relationship between these two organisms includes both **predation** and **mutual cooperation** (since they cooperate at least once a year :yum:). The population sizes of the two groups are denoted by x and h , and their relationship is governed by the following:

1. The growth rate of each population is proportional to its own population size, with proportionality constants a and b , respectively.
2. The growth rate of x is linearly related to h with a slope of $-c$.
3. The growth rate of h is linearly related to x with a slope of $-d$.

Here, a, b, c, d are constants.

Please derive and solve this system of equations. Also, indicate the conditions under which this system might collapse. (You cannot expect a population size to be negative or have infinite organisms on the Qingqing grassland.)

$$1. \frac{dx}{dt} = ax \quad \frac{dh}{dt} = bh$$

$$2. \frac{dx}{dt} = ax - ch \quad - \textcircled{1}$$

$$3. \frac{dh}{dt} = bh - dx \quad - \textcircled{2}$$

To find equilibrium points of the system, we set $\frac{dx}{dt}$ and $\frac{dh}{dt}$ to be 0.

$$\frac{dx}{dt} = ax - ch = 0$$

$$\frac{dh}{dt} = bh - dx = 0$$

$$ax = ch$$

$$x = \frac{ch}{a}$$

$$bh - d\left(\frac{ch}{a}\right) = 0$$

$$h\left(b - \frac{dc}{a}\right) = 0$$

$\therefore$ system in equilibrium if $b = \frac{dc}{a}$ or $h = 0$